

Native DS-TWR round-boundary investigation

Why shorter frames lost integrity, and why the guarded 53 ms profile works

Test date: July 31, 2026, EEST
Platform: 5 ESP32-S3 + DW3000 modules, Raspberry Pi dashboard
Branch: `experiment/native-ds-round-boundary`
Reference: `c727a64` before boundary instrumentation
Configuration: M1 active tag; M2, M3, M4, M5 anchors
Geometry: unchanged static indoor setup
Captures: five uninterrupted one-hour blocks

Key result. The apparent speed/precision limit was not the airtime of an ordinary POLL-RESP-FINAL exchange. All **267** impossible ToFs observed before the fix occurred at the separately inserted anchor-geometry boundary; **none** occurred in an ordinary tag ranging round. A 5 ms guard on each side of that periodic exchange eliminated every source rejection during a new one-hour run at the fastest tested **53 ms** profile. Delivered throughput changed only from **67.24** to **65.35 ranges/s** (−2.8%), while RESP timeouts fell by **53.7%** and FINAL timeouts by **96.6%**.

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1 Executive summary

The original hypothesis was that shorter Native DS-TWR profiles reduced accuracy because an individual radio exchange no longer fit inside its slot. The instrumentation added for this study shows a more specific failure: ordinary tag exchanges were sound, but the periodic transition into and out of anchor self-localization was not isolated from adjacent radio work.

Five one-hour captures were collected. Four unguarded profiles varied the normal frame from 59 ms down to 53 ms. The most aggressive 53 ms profile was then repeated after moving the normal gap directly behind the tag round and adding a 5 ms guard before and after the geometry exchange.

Table 1: Five-hour result summary. Rates and event counts come directly from the lossless firmware log stream.

Normal frame	Valid ranges	Rate	RESP timeout	FINAL timeout	Source reject
59 ms	220,710	61.31/s	4,657	350	34
58 ms	223,776	62.16/s	4,972	376	35
57 ms	226,315	62.86/s	6,660	858	98
53 ms	242,069	67.24/s	8,514	923	100
53 ms guarded	235,249	65.35/s	3,944	31	0

Decision. The guarded 53 ms profile is a valid fast operating point for the tested static setup. It preserves the short ordinary frame while spending extra time only around the periodic geometry-maintenance operation. The validation is strong evidence of timing integrity, but not yet a universal RF or motion qualification; dynamic, NLOS, and longer-duration trials should follow.

2 Question under test

The Native DS-TWR tag exchanges three mandatory packets with each anchor: POLL, RESP, and FINAL. With four anchors, one normal round contains four independent three-message exchanges. A separate anchor-to-anchor DS-TWR operation is periodically inserted so the dashboard can reconstruct a moving physical anchor geometry.

The investigation separated three possible causes:

1. **insufficient ordinary slot duration:** the packet exchange itself overruns its allocation;
2. **anchor-specific behavior:** A2, its RF path, or its antenna calibration is worse than the other anchors;
3. **round-boundary state overlap:** timestamps or radio state from the tag round and the separately inserted geometry exchange touch each other at the schedule boundary.

The third hypothesis is the only one consistent with all captured evidence.

3 Instrumentation and integrity checks

The firmware was instrumented at the radio boundary rather than only in the dashboard solver. Every exchange carries and validates:

- random token, 16-bit sequence, round, slot, and first-anchor ID;
- flags identifying a geometry exchange and the first following tag round;
- DW3000 transmit/receive timestamps for all stages;
- low-32 ESP timer values for host-side scheduling correlation;

- separate alternating initiator and responder exchange contexts.

Before applying the DS-TWR equation, the code requires exact context equality and valid timestamp intervals. A non-finite, non-positive, negative, or physically impossible ToF is rejected at the source before telemetry and position solving. Consequently, a clean trail cannot hide an invalid raw measurement behind later solver rejection.

The first anchor also rotates between rounds. If the failure remained tied to A2, that would indicate an anchor path. If it followed schedule position and appeared on all anchors, it would indicate a boundary problem.

4 Results before the guard

4.1 Throughput

Shortening the frame increases delivered ranges as expected. The pre-guard 53 ms profile reaches 67.24 valid ranges/s; the guarded validation reaches 65.35 ranges/s. The small difference is the cost of periodic boundary isolation, not an increase in the ordinary tag frame.

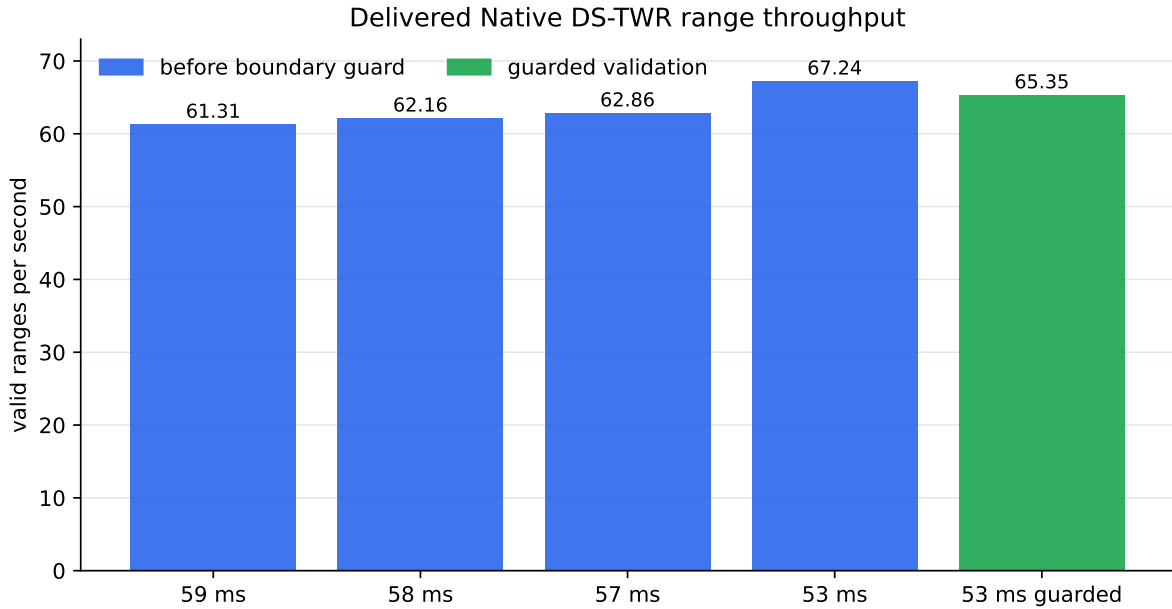


Figure 1: Delivered raw-range throughput. Blue denotes the original schedule; green denotes the corrected 53 ms validation.

4.2 Every impossible ToF is a boundary event

The decisive observation is categorical rather than statistical. Across the four original one-hour runs, all 267 source rejects occurred either inside the geometry exchange or in the first tag round immediately after it. There were zero source rejects in ordinary rounds.

Table 2: Location of every rejected ToF before and after the guard.

Frame	Geometry exchange	First post-geometry round	Ordinary round	Total
59 ms	29	5	0	34
58 ms	25	10	0	35
57 ms	81	17	0	98
53 ms	78	22	0	100
53 ms guarded	0	0	0	0

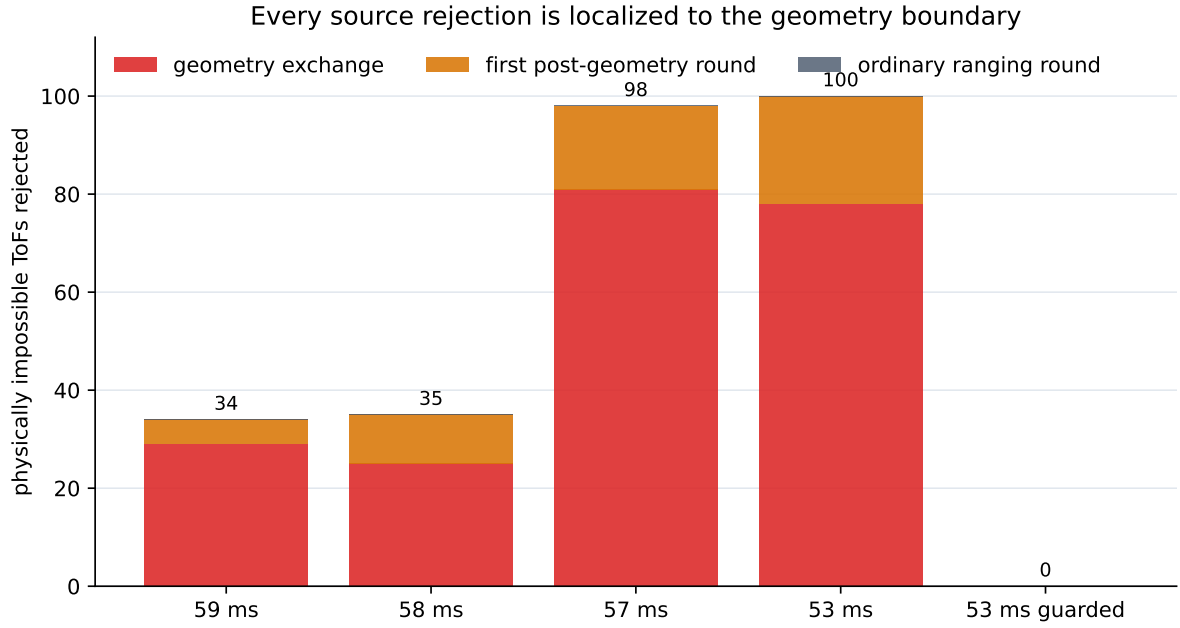


Figure 2: Source-rejected negative ToFs classified by scheduler location. The absence of an ordinary-round segment is the key result.

4.3 The problem is not A2

The rejected measurements appear on A2, A3, A4, and A5. Their distribution changes when the first-anchor order rotates. A2 has the fewest aggregated rejects, while A5 has the most; the fault therefore follows timing position, not one permanent anchor identity.

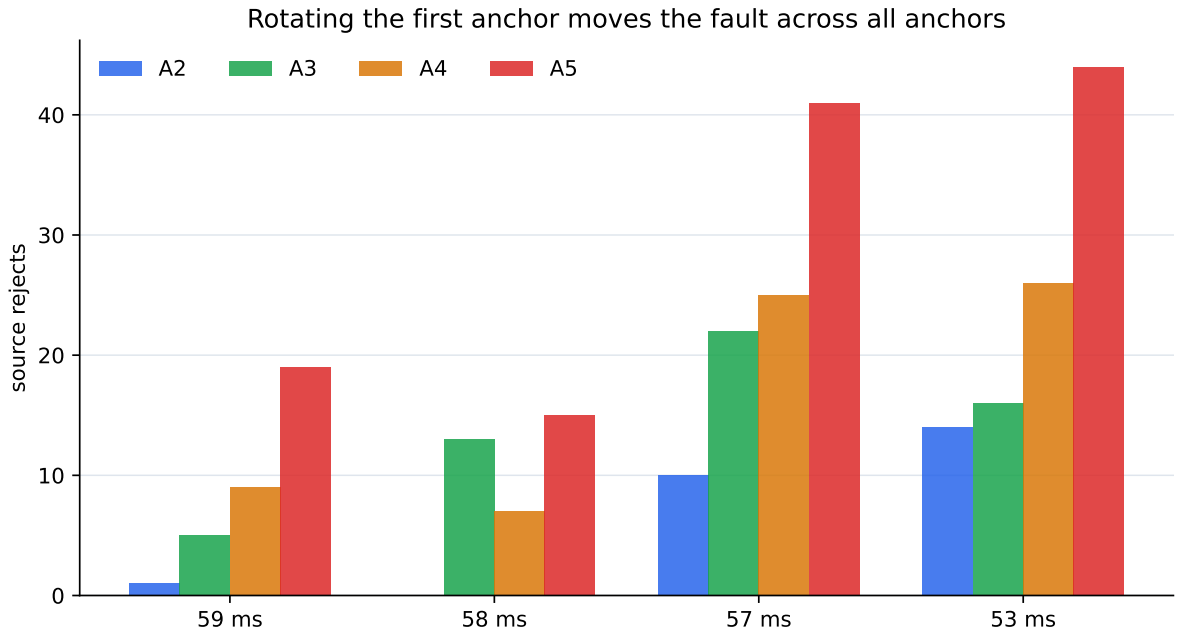


Figure 3: Rejected measurements by responding anchor. Rotation moves the boundary exposure across the four modules and disproves an A2-only fault.

5 Root cause

In the original scheduler, the configured round gap was applied after the optional geometry operation. It therefore did not isolate the last tag `FINAL` from the following `SURVEY_CMD`. On the other side of the

geometry slot, the next tag POLL relied only on the nominal slot duration. Occasional radio and host scheduling latency allowed the two initiator regimes to touch at either boundary.

This explains why packet-airtime arithmetic alone was insufficient. An ordinary three-message exchange could fit comfortably while the separately inserted maintenance exchange still collided with the state transition around it. The resulting failure was rare, short, and often invisible after solver gating, but it was present in the raw timestamps.

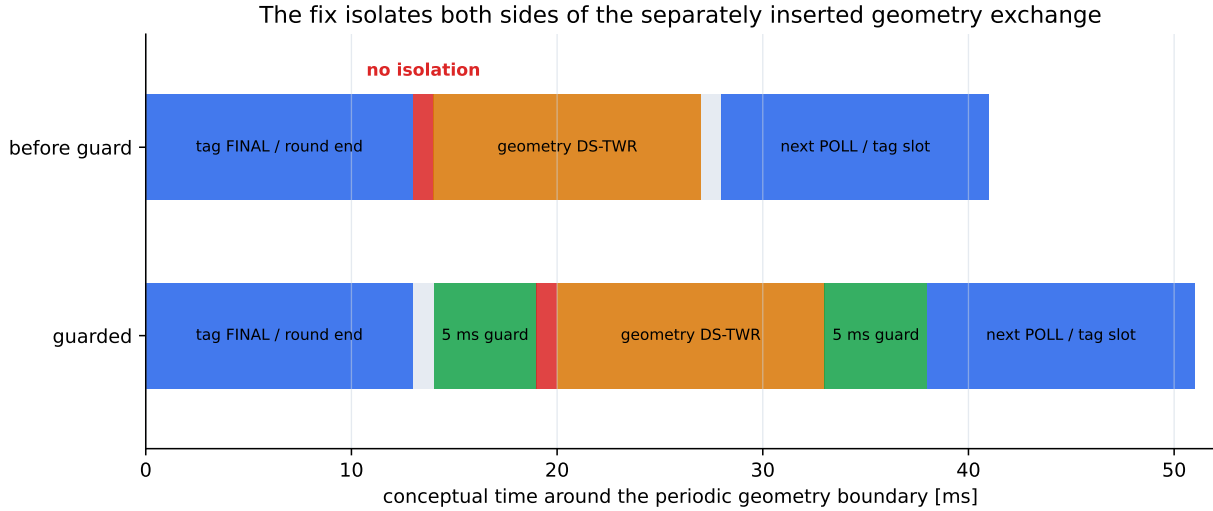


Figure 4: Conceptual scheduler comparison. The correction protects both sides of geometry maintenance without lengthening an ordinary 53 ms tag round.

6 Correction

The corrected schedule makes five changes:

1. place the configured frame gap immediately after every tag round;
2. wait 5 ms before transmitting `SURVEY_CMD`;
3. wait another 5 ms after the anchor-geometry DS-TWR slot;
4. mark every slot in the first following tag round as post-geometry;
5. retain complete diagnostic traces for every rejected exchange while sampling detailed accepted traces to avoid perturbing the scheduler.

The collector was also corrected to deduplicate by token, round, and slot rather than by the wrapping 16-bit sequence alone. The final one-hour run crossed several sequence wraps while its count remained linear.

7 Guarded 53 ms validation

The final firmware was rebuilt, uploaded simultaneously to all five modules, and tested for one uninterrupted hour at slot 13 ms plus gap 1 ms. It produced 235,249 accepted ranges, no impossible ToF, no timestamp-context mismatch, and no ordinary-round integrity error.

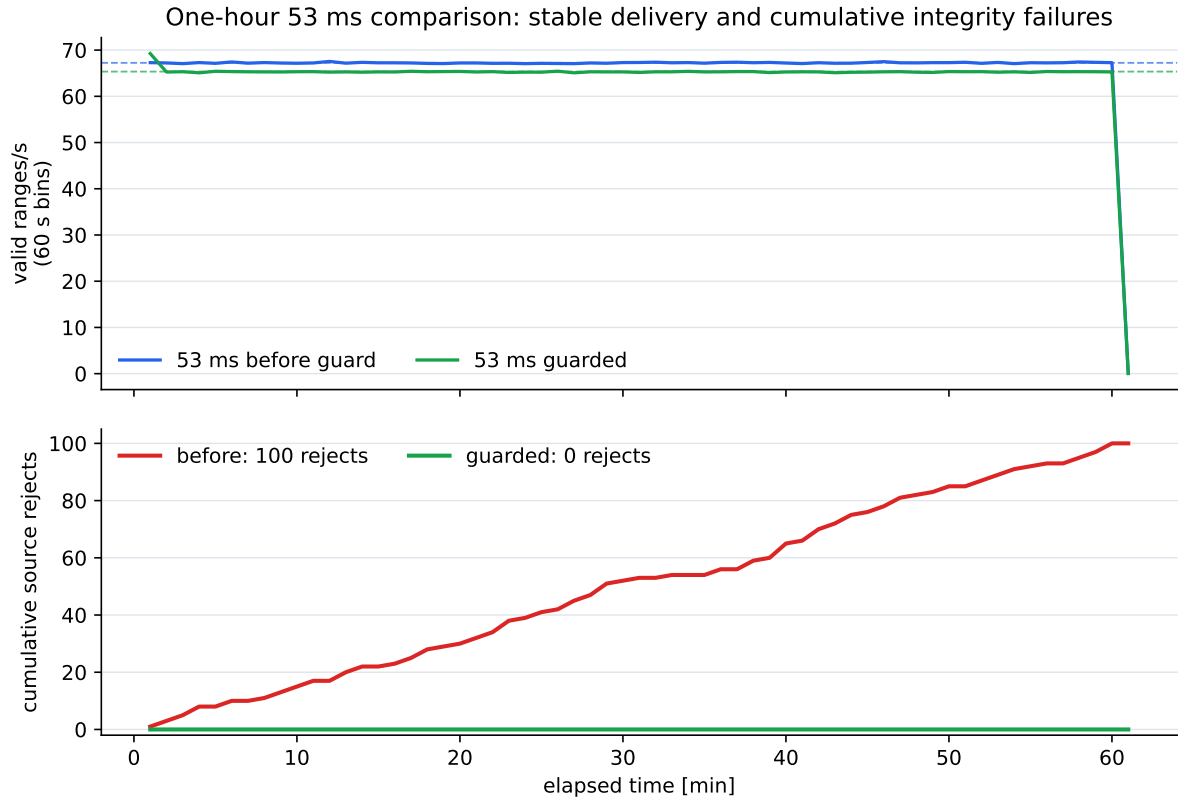


Figure 5: Per-minute delivery and cumulative source rejects for the two 53 ms runs. The guarded capture remains at zero rejects for the full hour.

The guard reduces valid throughput by 2.8%, from 67.24 to 65.35 ranges/s. That cost is smaller than the nominal added guard time would suggest because fewer collided exchanges are retried or lost. RESP timeouts fall from 8,514 to 3,944 (−53.7%), FINAL timeouts fall from 923 to 31 (−96.6%), and source-rejected ToFs fall from 100 to zero.

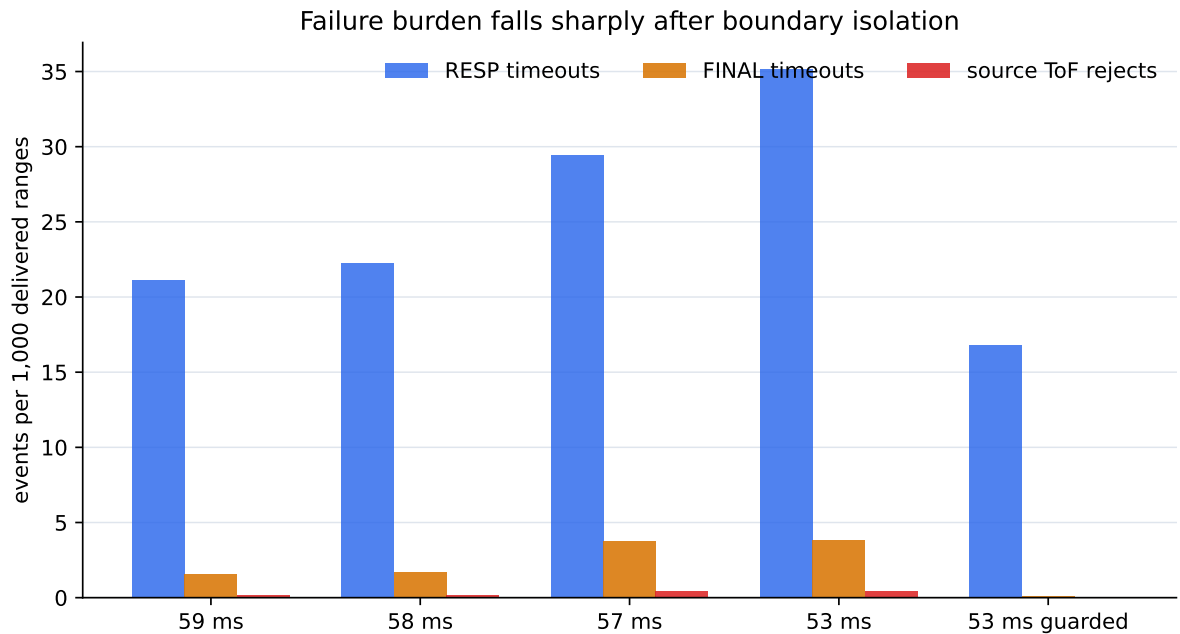


Figure 6: Failure events normalized per 1,000 delivered ranges. The guarded schedule improves both integrity and radio-exchange completion.

8 How to interpret the result

8.1 What is demonstrated

- The ordinary 53 ms Native DS-TWR frame is not the source of the observed impossible ranges in this static test.
- The prior apparent accuracy limit was a periodic scheduler-boundary problem, not a fundamental DS-TWR propagation or airtime limit.
- The error is not specific to A2, antenna delay, or one RF path.
- Source validation prevents a rare timestamp corruption from becoming a plausible-looking distance or position.
- The boundary guards buy clean integrity for only 2.8% measured delivered-throughput cost.

8.2 What is not yet demonstrated

This test does not establish a universal 53 ms limit. It uses one static indoor geometry and one RF environment. A clean source-reject count proves timestamp and schedule integrity, but it is not by itself a position-accuracy metric. The next qualification should include moving-tag trials, controlled NLOS conditions, power/temperature variation, and multiple runs of at least one hour. Position RMSE and P95 should be compared against the same fixed geometric reference used for the slow control.

8.3 Recommended operating policy

Use the guarded 53 ms profile as the current fast candidate. Keep the 59 ms and 58 ms profiles available as diagnostic controls, but the present evidence does not justify slowing ordinary tag rounds merely to hide a geometry boundary that can be isolated directly. Preserve all source context checks and physically impossible-ToF rejection in production builds.

9 Reproducibility and artifacts

The complete raw captures, metadata snapshots, generated figures, and machine-readable report summary live under:

```
reports/native_ds_round_boundary_20260731/
```

The classification can be reproduced with:

```
tools/analyze_native_ds_round_boundary.py \
  reports/native_ds_round_boundary_20260731/data/\
candidate_59ms_round_boundary_3600s.logs.jsonl \
  reports/native_ds_round_boundary_20260731/data/\
candidate_58ms_round_boundary_3600s.logs.jsonl \
  reports/native_ds_round_boundary_20260731/data/\
candidate_57ms_round_boundary_3600s.logs.jsonl \
  reports/native_ds_round_boundary_20260731/data/\
candidate_53ms_round_boundary_3600s.logs.jsonl \
  reports/native_ds_round_boundary_20260731/data/\
candidate_53ms_guarded_3600s.logs.jsonl
```

Figures and `report_summary.json` are regenerated by:

```
tools/uwb_native_ds_round_boundary_report.py
```

The interrupted 130 s over-instrumented capture is retained separately and is not included in any result above.